#define [citation needed?] I am 90% sure this is good enough, but if this needs more elaboration on a test, please let me know otherwise you can ignore this comment.

5 hours on examples

Problem 1 (17AMO1)

For any prime $p \equiv 1 \bmod 4$, we claim that $a = \frac{p+1}{2}; b = \frac{3p-1}{2}$ works. Note that,

$$\gcd(a, b) = \gcd\left(\frac{p+1}{2}, \frac{3p-1}{2}\right) = \gcd\left(\frac{p+1}{2}, \frac{p-3}{2}\right) = \gcd\left(2, \frac{p-3}{2}\right)$$

Since $p \equiv 1 \bmod 4$, $\frac{p-3}{2}$ is odd so they are relatively prime. Note that,

$$a + b \mid a^b + b^a \Leftrightarrow a^b + b^a \equiv 0 \bmod a + b$$

$$\Leftrightarrow (-b)^b + b^a \equiv 0 \bmod a + b$$

$$\Leftrightarrow b^a \equiv b^b \bmod a + b$$

Since $\gcd(b, a + b) = 1$,

$$\Leftrightarrow b^{b-a} \equiv 1 \bmod a + b$$

We know $a + b = 2p$ and $b - a = p - 1$. Since $\phi(2p) = p - 1$, by Euler's Totient Theorem, since $\gcd(b, a + b) = 1$, this works.

By Dirichlet, there are infinitely many primes $p \equiv 1 \bmod 4$, so we are done.

Problem 2 (99SLN3)

We let $a_i = p_i^2$ where p_i is the i-th prime that is $1 \bmod 4$, of which there are infinitely many by Dirichlet. Since they are relatively prime, we independently check that there exists a b_n that is divisible by both $a_n, a_n + 1$.

For,

$$a_n + 1 = p_i^2 + 1 \mid b_n^2 + 1$$

This is clearly possible for $b_n \equiv p_i \bmod p_i^2 + 1$. For,

$$p_i^2 \mid b_n^2 + 1$$

By (lemma I don't know), -1 is a quadratic residue mod p_i^2, so $b_n \equiv a \bmod p_i^2$ for some a. By CRT, we have, for some b,

$$b_n \equiv b \bmod p_i^2\left(p_i^2 + 1\right)$$

We may then find a sufficiently large $b_n > b_{n-1}$.

Problem 3 (19USMCA1)

We claim that Kelvin may append either a 7 or 8 to the end of the string for every move of the game such that Alex cannot have a move that wins on the next round.

For the first move, Kelvin plays a 7, which is not a square. Alex obviously cannot win on the next move.

For every move after, Kelvin appends either a 7 or 8. Since $7, 8$ are not quadratic residues mod 10, the current number is not a square. For the same reason, for Alex to win on the next move, he must append a number to the end of the string.

At this point, there will be 3 numbers on the board after Kelvin and Alex move next with the first being 7. Therefore, it is the square of at least 25, so Alex cannot have a winning response for both 7 and 8 being appended.

6 clubs; 8 hours

Problem 5 (09APMO4)

The case where $n = 1, 2$ is trivial.

For $n > 2$, we may choose the fractions with value,

$$\frac{n!^2 + 1}{n!}, \frac{n!^2 + 2}{n!}, \dots, \frac{n!^2 + n}{n!}$$

Note that $\gcd(n! + k, n!) = \gcd(k, n!) = k$ for all $1 \leq k \leq n$. Therefore, these respectively simplify to,

$$a_i = \frac{n!^2}{i} + 1; b_i = \frac{n!}{i}$$

We have,

$$a_1 > a_2 > \cdots > a_n = n!\,(n-1)! + 1 > n! = b_1 > b_2 > \cdots > b_n$$

Therefore, all $2n$ numbers are pairwise distinct as desired.

Problem 7 (23USEMO1)

Couldn't do this at all in the contest and solved this today in 20 min lol.

Define a_b to be a base b number.

Lemma: For a pair of integers a, b, n, m, where $k = \gcd(n, m)$, we may find infinitely integers x that satisfy $x \equiv a \bmod n$ and $x \equiv b \bmod m$ if $a \equiv b \bmod k$.

Proof: Same as proof of CRT.

Call a number k-beautiful for some $k \geq 4$ if, for every $4 \leq b \leq k$, the base-b representation of the number has digits $2, 0, 2, 3$ in order.

We prove that, for all $k \geq 4$, we may find distinct numbers a, n such that all numbers of the form $a \bmod n$ are k-beautiful. We do this with induction.

Base case: $k = 4$

All numbers that are $2023_4 \bmod 10000_4$ satisfy this.

Inductive step: $k = x \to k = x + 1$

By the inductive hypothesis, for some a, n, we have all numbers that are $a \bmod n$ are x-beautiful.

Let y be a positive integer such that $(x+1)^y > n$. Clearly, all numbers which are,

$$(2023_{x+1}) \cdot (x+1)^y + m \pmod{(x+1)^{y+4}}$$

For $0 \leq m < (x+1)^y$ are $x+1$-beautiful since the $y+1, y+2, y+3, y+4$ digits from the right are 2023 in that order.

Note that $\gcd(n, (x+1)^y) \leq n < (x+1)^y$, therefore, by Pigeonhole, there exists a m on this range such that,

$$(2023_{x+1}) \cdot (x+1)^y + m \equiv a \pmod{\gcd(n, (x+1)^y)}$$

Using this selection of m, we apply the lemma to get for some a', n', all numbers that are $a' \bmod n'$ are still x-beautiful and now additionally $(x+1)$-beautiful.

Conclusion: By mathematical induction, we are done.

Hence, there exists an infinite number of 10000-beautiful numbers as desired.

17 clubs; 9.5 hours

<u>Problem 10 (87IMO5)</u>

We start with the lattice points, (x, x^2) for integers x in range $0 \leq x \leq 1986$. Namely, the points,

$$(0,0); (1,1); (2,4); (3,9); \dots; (1986, 1986^2)$$

Since the graph of $f(x) = x^2$ is convex, its derivative is monotonic increasing. Therefore, no three distinct points on the graph may be collinear *(by mean value theorem)*.

Therefore, by shoelace, the current area of every triangle is a positive rational.

We may use the distance formula to write the distance between every pair of points as a not-necessarily-simplified radical $\sqrt{n}$. Note that the distance is maximized between $(0,0)$ and $(1986, 1986^2)$, so,

$$n \leq 1986^2 + 1986^4$$

To get our construction, we perform a dilation about the origin with scale $\sqrt{p}$ for some prime $p > 1986^2 + 1986^4$.

The area of every triangle is now $\left(\sqrt{p}\right)^2 = p$ times what it used to be, so still rational.

The distance between every pair of points may now be expressed as $\sqrt{np}$. Since $p > n > 0$, we cannot have $p \mid n$. Therefore, $\sqrt{np}$ must be irrational. [citation needed?]

Problem 15 (07USATST4)

We claim the answer is no.

Claim: For any fixed integers (a, b), the sequence $b, b^b, b^{b^b}, \ldots$ will become constant modulo a.

Note that for a sufficiently large x, $b^{x+\phi(b)} = b^x \bmod a$. Therefore, we may prove that $b, b^b, b^{b^b}, \ldots$ will become constant modulo $\phi(a)$. We use induction to get eventually we have $\phi\left(\phi(\ldots\phi(a))\right) = 1$, by which time we'll be done.

Problem 17 (18EGMO2)

For any odd n, $3 \mid 2^n + 1$. Therefore, we may choose $x = 3; y = \frac{2^n+1}{3}$ for any $n > 3$, of which there are infinitely many,

Note that, applying $1 + \frac{1}{2^n}$ then applying the 2 multiplication n times.

$$f(2^n + 1) \leq f(2^n) + 1 \leq n + 1$$

Additionally, we also obviously have,

$$f(x) = f(3) = 2$$

Claim: $f\left(\frac{2^n+1}{3}\right) \geq n$

FTSOC, let this be attainable in $n - 1$ moves or less. Note that we must choose 2 at least $n - 2$ times for if not, the maximum we can attain is obviously,

$$2^{n-3} \cdot \left(\frac{3}{2}\right)^2 < \frac{2^n + 1}{3}$$

$$\Leftrightarrow 2^{n-3} \cdot \frac{27}{4} < 2^n + 1 \quad (!)$$

Therefore, for some x,

$$\frac{2^n + 1}{3} = 2^{n-2} \cdot \left(\frac{x+1}{x}\right)$$

However, since $2^n + 1$ is odd, the RHS must also be an odd integer. The only way this is possible is if $v_2(x) = n - 2$. Since $x \leq 2^{n-2}$, we have $x = 2^{n-2}$. Therefore,

$$\frac{2^n + 1}{3} = 2^{n-2} + 1$$

$$\Leftrightarrow 2^n = 3 \cdot 2^{n-2} + 2$$

$$\Leftrightarrow 2^{n-2} = 2$$

For $n > 3$, this is clearly impossible.

With the claim proven, we hence have, for this selection of x, y, of which there are infinitely many,

$$f(xy) \leq n + 1 < f(x) + f(y)$$

26 clubs; 13 hours

<u>Problem 18 (15BRA3)</u>

(I got the 27, 169 by myself :D but couldn't figure out square-free)

We claim all n of the form,

$$n = 27(169 \cdot 78t - 25)$$

Where $169 \cdot 78t - 25$ and $27 \cdot 78t - 4$ are square-free numbers work.

Note that, for any square free integer y, we obviously have $f(y) = 1$. Additionally, f is obviously multiplicative [citation needed?]. Since $169 \cdot 78t - 25$ is square free and relatively prime to 3 and hence 27,

$$f(n) = f(27) \cdot f(169 \cdot 78t - 25) = f(3^3) = 27$$

On the other hand,

$$f(n-1) = f(27 \cdot 169 \cdot 78t - 27 \cdot 25 - 1) = f\big(169(27 \cdot 78t - 4)\big)$$

Clearly, $27 \cdot 78t - 4$ is relatively prime to 13 and hence 169. Therefore,

$$= f(169) \cdot f(27 \cdot 78t - 4) = 26$$

Therefore, $f(n) = f(n-1) + 1$ as desired.

We now check that there are infinitely many t which satisfy this condition that $169 \cdot 78t - 25$ and $27 \cdot 78t - 4$ are both square-free.

Claim: For a fixed number M, there are at most $\sum_{p=5;p \text{ is prime}}^{\sqrt{\max(169 \cdot 78M-2\ \ ,27 \cdot 78M-4)}} \left(2\left\lceil\frac{M}{p^2}\right\rceil\right)$ number of integers t on $1 \leq t \leq M$ such that at least one of $27 \cdot 78t - 4$ and $169 \cdot 78t - 25$ is not square-free.

Consider what the squares primes may divide $27 \cdot 78t - 4$ or $169 \cdot 78t - 25$. Obviously, if,

$$p^2 > \max(27 \cdot 78t - 4, 169 \cdot 78t - 25)$$

This is impossible. For all other primes p, we'd want $27 \cdot 78t \equiv 4 \bmod p^2$ or $169 \cdot 78t \equiv 25 \bmod p^2$. This is impossible if $p = 13$ or $p = 3$. For all other p, we may respectively take modular inverse and simplify that to $t \equiv x \bmod p^2$ and $t \equiv y \bmod p^2$ respectively for some not-necessarily-distinct x, y.

Since we are only taking the upper bound, we are not concerned about overcounting, which would reduce the actual maximum. Therefore, we have at most $2\left\lceil\frac{M}{p^2}\right\rceil$ for this prime. Using the same argument, we may just sum these values over all the primes from 5 to $\sqrt{\max(169 \cdot 78M - 25, 27 \cdot 78M - 4)}$. [citation needed?]

However,

$$\sum_{p=5;p\text{ is prime}}^{\sqrt{\max(169\cdot 78M-25,27\cdot 78M-4)}} \left(2\left\lceil\frac{M}{p^2}\right\rceil\right)$$

$$< \sqrt{\max(169 \cdot 78M - 25, 27 \cdot 78M - 4)} + \sum_{p=5;p\text{ is prime}}^{\infty} \left(\frac{2M}{p^2}\right)$$

$$< \sqrt{\max(169 \cdot 78M - 25, 27 \cdot 78M - 4)} + 2M \sum_{p=5}^{\infty} \left(\frac{1}{p^2}\right)$$

$$= \sqrt{\max(169 \cdot 78M - 25, 27 \cdot 78M - 4)} + 2M \cdot \left(\frac{\pi^2}{6} - 1 - \frac{1}{4} - \frac{1}{9} - \frac{1}{16}\right)$$

$$< \sqrt{\max(169 \cdot 78M - 25, 27 \cdot 78M - 4)} + \frac{M}{2}$$

Therefore, using complementary counting, the amount of t such that they are both square-free.

$$M - \sum_{p=5;p\text{ is prime}}^{\sqrt{\max(169\cdot 78M- \quad ,27\cdot 78M-4)}} \left(2\left\lceil\frac{M}{p^2}\right\rceil\right) > \frac{M}{2} - \sqrt{\max(169 \cdot 78M - 25, 27 \cdot 78M - 4)}$$

Taking M large, this will approach infinity.

<u>Problem 23 (13SLN4)</u>

We claim that this is impossible.

FTSOC, let there be such a sequence such that for some k, the numbers $\overline{a_k a_{k-1} \dots a_0}, \overline{a_{k+1} a_k \dots a_0}, \overline{a_{k+2} a_{k+1} \dots a_0}, \dots$ are all perfect squares. We let,

$$x_k^2 = \overline{a_k a_{k-1} \dots a_0};$$

$$x_{k+1}^2 = \overline{a_{k+1} a_k \dots a_0};$$

$$\dots$$

We hence have,

$$x_{k+1}^2 - x_k^2 = (x_{k+1} - x_k)(x_{k+1} + x_k) = a_{k+1} \cdot 10^{k+1};$$
$$x_{k+2}^2 - x_{k+1}^2 = (x_{k+2} - x_{k+1})(x_{k+2} + x_{k+1}) = a_{k+2} \cdot 10^{k+2};$$
$$\dots$$

Case 1: $v_5(x_k^2) = 2e < k + 1$

In that case, for $v_5(x_{k+1}^2 - x_k^2) \geq k + 1 > v_5(x_k^2)$, we must have $v_5(x_{k+1}^2) = 2e$ too. Inducting, we have $2e = v_5(x_k^2) = v_5(x_{k+1}^2) = v_5(x_{k+2}^2) = \cdots$.

Therefore, since exactly one may be more than e,

$$\min\big(v_5(x_{k+1} - x_k), v_5(x_{k+1} + x_k)\big) = e;$$

We then have,

$$\max\big(v_5(x_{k+1} - x_k), v_5(x_{k+1} + x_k)\big) = k + 1 - e;$$

Therefore, (I was following solutions and I don't understand the logic of this part, I've asked in OTIS Discord, but nobody has answered yet. Wouldn't it be $\max(x_{k+1} - x_k, x_{k+1} + x_k) \geq 5^{k+1-e}$?)

$$\min(x_{k+1} - x_k, x_{k+1} + x_k) \geq 5^{k+1-e}$$

For sufficiently large k, we have this is greater than $\sqrt{9 \cdot 10^{k+1}}$. Since $(x_{k+1} - x_k)(x_{k+1} + x_k) \leq 9 \cdot 10^{k+1}$, this is a contradiction.

Case 2: $v_5(x_k^2) \geq k + 1$

We similarly use induction to find $v_5(x_k^2) \geq k + 1$ for all sufficiently large k. Additionally, inducting backwards gives $v_5(x_k^2) \geq k + 1$ for all k. Therefore, $v_5(x_0^2) \geq 1$, which is possible only if $x_0 = 25$. Thus, all x_k are odd.

$$(x_{k+1} - x_k)(x_{k+1} + x_k) = a_{k+1} \cdot 10^{k+1}$$

Since x_k is odd, $\min\big(v_2(x_{k+1} - x_k), v_2(x_{k+1} + x_k)\big) = 1$ so one is divisible by 2^k. Therefore,

$$\min(x_{k+1} - x_k, x_{k+1} + x_k) \geq 2^k \cdot 5^{\frac{k+1}{2}}$$

Once again, this is larger than $\sqrt{9 \cdot 10^k}$ for sufficiently large k.

Both cases give contradictions, so we are done.

40 clubs; 17 hours